$$f(x_1, x_2) = \sin\left(\begin{matrix}5x_1 - 5\\ 2\end{matrix}\right)\cos(2.5 - 5x_2) + \frac{1}{10}\left(\begin{matrix}5x_2 + 1\\ 2\end{matrix}\right)^2$$

This function simulates the accuracy surface of a Support Vector Machine (SVM) model in a hyperparameter tuning task. The *x*-axis denotes the value of the penalty parameter *C*, while the *y*-axis denotes the value for the RBF kernel parameter . (We will be using this function as our objective in future chapters as well.)

2. Visualize the function over the domain *[0, 2]^2*. The heat map should look like Figure A.2.

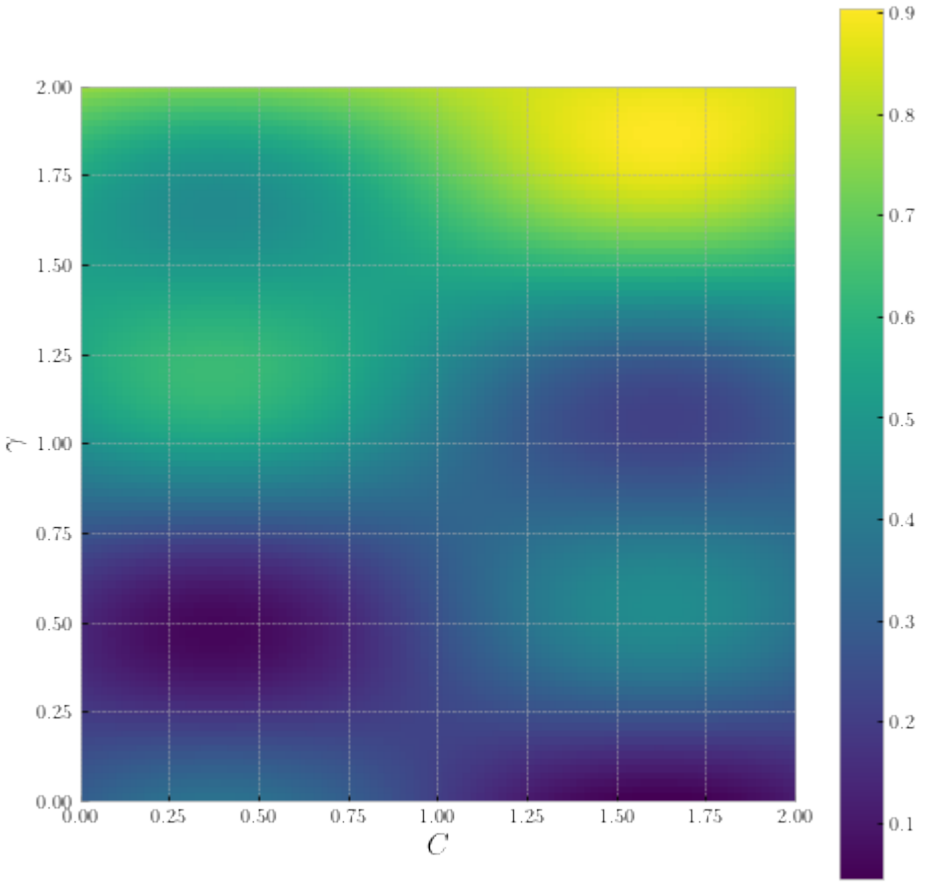


Figure A.2 Accuracy of an SVM model on a test dataset, as a function of the penalty parameter C and the RBF kernel parameter . The function changes more quickly with respect to than to C.

3. Randomly draw 100 data points from the domain *[0, 2]^2*. This will be used as our training data.
4. Implement a GP model with a constant mean function and a Matérn 2.5 kernel with an output scale implemented as a `gpytorch.kernels.ScaleKernel` object.
5. Don't specify the `ard_num_dims` parameter when initializing the kernel object, or set the parameter to `None`. This will create a GP model without ARD.
6. Train the hyperparameters of the GP model using gradient descent, and inspect the length scale after training.
7. Redefine the GP model class, this time setting `ard_num_dims = 2`. Retraining the GP model with gradient descent, and verify that the two length scales have significantly different values.

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